Translationese All the Way Down

Auditing what "supports Twi" means in social media NLP resources

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Repository: https://github.com/osamabinIaggin/dcit316-final-project

Abstract

Google Translate, Meta's NLLB-200 and three human-annotated benchmarks list Twi as a supported language. This project asks what that support is worth for Twi as it is written on social media. I audited four resources against six documentation criteria, measured how far the three existing Twi corpora sit from tweet text, and evaluated five sentiment systems on the 949-tweet AfriSenti Twi test split. None of the resources names a Twi dialect; FLORES-200 holds two independent Twi translations under the codes `aka_Latn` and `twi_Latn`; and both benchmarks behind Twi capability claims are translations into Twi. The tweets differ sharply from them: 42.5% write `3` in place of ɛ, against 0.1% or fewer of benchmark sentences. On the test split, a TF-IDF and logistic regression baseline trained on Twi reaches macro-F1 0.593. An English Twitter model reaches 0.243, XLM-T 0.337, and NLLB translation followed by English classification 0.373 (all $p < 0.001$ against the baseline). The off-the-shelf systems fail by predicting neutral: the English model labels 83.3% of positive and negative tweets as neutral. NLLB passes `3`-spellings through untranslated in 48.6% of affected tweets, and declaring the source as `aka_Latn` instead of `twi_Latn` changes the final label for 25.6% of tweets. For Ghanaian social media, a little in-domain labelled data beats every system that claims broader coverage.

1. Introduction

Google Translate added Twi and Ewe in 2022. Meta's NLLB-200 translates both, and its evaluation set, FLORES-200, carries files for `twi_Latn`, `aka_Latn` and `ewe_Latn`. On the research side, AfriSenti (Muhammad et al., 2023a), AfriHate (Muhammad et al., 2025) and MasakhaNER 2.0 (Adelani et al., 2022a) all list Twi among their languages. A student in Accra who wants to analyse Ghanaian online discourse could fairly conclude that the tools exist.

I wanted to know what that support amounts to for the text people in Ghana actually post. Twi on social media is short, informal and often mixed with English, and it is often typed without the letters ɛ and ɔ. AfriSenti's authors put this down to keyboards that lack those letters, and note that ɛ is typed as `3` and ɔ as `c`. Benchmarks rarely look like this.

A claim that a system "supports Twi" bundles two different things. One is capability: does the system work on the text you have? The other is accountability: can someone outside the project find out which Twi was covered, where the text came from, who labelled it and how reliably? The second can be checked from documents; the first needs experiments. This project does both.

Research questions.

- **RQ1 (accountability).** What do the public resources that claim Twi coverage disclose about the language variety, the provenance of the text, the annotators and their agreement?

- **RQ2 (distribution).** How far is the text these claims are measured on from Twi as it is written on social media?
- **RQ3 (capability).** How do the tools a Ghanaian analyst would reach for perform on natively written Twi tweets, compared with a simple baseline trained on Twi?

Contributions. A documentation audit of four resources against six criteria, two of which (provenance of the text, and comparability of agreement scores) are added here; a measurement of the distance between the three Twi corpora that exist; and an evaluation of five sentiment systems on AfriSenti Twi, with an error analysis by orthography, English mixing, length and emoji. All code, saved predictions, metrics and executed notebooks are in the repository above.

2. Related work

African-language social media data. AfriSenti (Muhammad et al., 2023a) and its shared task (Muhammad et al., 2023b) released sentiment-labelled tweets for 14 African languages; the Twi subset is the only human-labelled sentiment data for a Ghanaian language I could find. AfriHate (Muhammad et al., 2025) adds hate and abusive-language labels for 15 languages including Twi. MasakhaNER 2.0 (Adelani et al., 2022a) annotates named entities in 20 languages including Twi and Ewe, and draws the text for several of them from MAFAND-MT (Adelani et al., 2022b), a corpus of French and English news translated into African languages.

Translation benchmarks and their quality. FLORES-200 was built for NLLB-200 (NLLB Team et al., 2022) by translating 2,009 English Wikimedia sentences into roughly 200 languages. Abdulmumin et al. (2024) had native speakers review the FLORES files for Hausa, Northern Sotho, Xitsonga and isiZulu and found enough errors to release corrections. No Ghanaian language was covered.

Language identification. Caswell et al. (2020) showed that identifiers which score well on held-out sets can produce web corpora in which most text labelled as a low-resource language is something else. Burchell et al. (2024) found that identifying languages in code-switched text remains unsolved. Both matter here because AfriSenti assigned most of its Twi tweets to Twi with a model trained on 2,000 hand-labelled examples.

Sentiment models for tweets. The English model evaluated here comes from the TimeLMs series (Loureiro et al., 2022), fine-tuned on the TweetEval sentiment task (Barbieri et al., 2020). XLM-T (Barbieri et al., 2022) continues training XLM-R on multilingual tweets and fine-tunes it for sentiment in eight languages, none of them African.

Documentation standards. Data statements (Bender and Friedman, 2018), datasheets (Gebru et al., 2021) and model cards (Mitchell et al., 2019) set out what dataset and model creators should record. The audit criteria in Section 4.1 build on them.

The gap this project addresses: the African NLP literature reports what systems achieve on its benchmarks, and the documentation literature says what should be disclosed, but I found no work that checks Ghanaian-language claims against both the documentation and the text Ghanaians write.

3. Data and ethics

Source	What is used	Size	Natively written?
AfriSenti Twi, commit `5aec3cf`	Tweets with sentiment labels	3,481 train, 388 dev, 949 test	Yes
MasakhaNER 2.0 Twi, commit `ba5843c`	Sentences only, no entity labels	6,056 sentences	No (MAFAND-MT translation)
FLORES-200 `twi_Latn`, `aka_Latn`, `eng_Latn`	dev + devtest	2,009 sentences each	No (translated from English)
Four papers (AfriSenti, AfriHate, MasakhaNER 2.0, FLORES correction)	Text extracted from the PDFs	n/a	n/a

The AfriSenti Twi test split holds 450 positive, 353 negative and 146 neutral tweets. Its reported inter-annotator agreement is 0.51 (free-marginal kappa).

Everything is public. `scripts/fetch_data.py` downloads each file from a pinned commit and records a SHA-256 checksum, so no raw data is committed; the AfriSenti and MasakhaNER repositories declare no licence, and I did not want to redistribute text I had no explicit permission to share. AfriSenti tweets carry no user handles or links (none of the 4,818 tweets contains one). No new data was collected and no person was contacted. The few tweets quoted in Section 7 are already distributed in a public benchmark and identify nobody.

I do not read Twi well enough to judge translation quality, so no result depends on my reading of Twi. Where a native speaker's judgement is needed, the paper says so. The full data card, ethics statement and AI-use disclosure are in the repository.

4. Methods

4.1 Documentation audit (RQ1)

Each paper is converted to text twice with `pdftotext`: once with `-layout`, which keeps tables intact, and once in reading order, because `-layout` interleaves the two columns of an ACL paper and breaks any sentence that wraps. `scripts/audit_documentation.py` searches both for patterns tied to six criteria and saves every matching line with its line number.

	Criterion	Basis
A1	Is the language referent precise (Akan, Twi, a named dialect)?	Bender and Friedman (2018)
A2	Is the text natively written or translated?	Added here
A3	Are native-speaker annotators stated?	Gebru et al. (2021)
A4	Is agreement reported, and in a form comparable across resources?	Added here
A5	Is annotator pay disclosed?	Gebru et al. (2021)
A6	Is there a datasheet, data statement or model card?	Gebru et al. (2021); Mitchell et al. (2019)

The search is automatic; each verdict in the matrix is my reading of the matches, and the matches are committed so the reading can be checked. "Not found" means the fact is absent from the public paper, not that the practice did not happen.

4.2 Domain ladder (RQ2)

The three Twi corpora differ in how far they sit from spontaneous writing: FLORES-200 is translated encyclopaedic prose, MasakhaNER 2.0 is translated news, and AfriSenti is tweets. For every text I compute the share of tokens using ɛ or ɔ, whether the text writes `3` inside a word in place of ɛ, the share of tokens containing letters outside the Twi alphabet (c, j, q, v, x, z), the share of emoji-bearing texts, and an English-token share. The English lexicon is FLORES English plus scikit-learn's stop-word list, minus every word that appears in FLORES Twi or Akan; this makes the estimate zero on FLORES by construction, so it is only compared across the other two corpora. I also measure how much of the tweets' vocabulary each benchmark covers, and how similar the FLORES `aka_Latn` and `twi_Latn` files are.

4.3 Sentiment systems (RQ3)

All systems are scored on the 949-tweet AfriSenti Twi test split, which is used once.

System	What it is	Role
Majority class	Most frequent training label (positive)	Floor
TF-IDF + logistic regression	Character 2–5-grams and word 1–2-grams, class-balanced, C chosen on dev	**Baseline**: trained on Twi
English-only	`cardiffnlp/twitter-roberta-base-sentiment-latest`, applied directly to Twi	What an analyst uses by default
XLM-T	`cardiffnlp/twitter-xlm-roberta-base-sentiment`, zero-shot	Multilingual, no Twi training
NLLB → English	`nllb-200-distilled-600M` translates Twi to English, then the English-only model	Translate-then-classify

The NLLB pipeline is run twice on identical input, once declaring the source as `twi_Latn` and once as `aka_Latn`, because FLORES treats these as separate languages and no resource says which one tweets belong to. Decoding is greedy with at most 96 new tokens.

Metrics. Macro-F1 is the primary metric because the classes are unbalanced; accuracy, weighted F1 and per-class F1 are also reported. A 95% bootstrap interval (1,000 resamples, seed 22128981) accompanies each macro-F1, and an exact McNemar test compares each system with the baseline. I also report the **silent-failure rate**: of the tweets whose gold label is positive or negative, the share predicted neutral. This error produces no warning, because a confident "neutral" looks like an ordinary answer.

Error analysis. The test split is cut by orthography (uses ɛ/ɔ, writes `3` for ɛ, neither), by English-token share, by length and by the presence of emoji, and macro-F1 is computed per slice. Samples of three error types are saved for inspection: sentiment lost through translation, tweets every non-trivial system gets wrong, and label changes caused only by the declared source language.

5. Results

5.1 Coverage

Language	FLORES-200	MasakhaNER 2.0	AfriSenti	AfriHate	Google Translate	NLLB-200
Twi / Akan	yes (two files)	yes	yes	yes	yes (2022)	yes
Ewe	yes	yes	no	no	yes (2022)	yes
Ga	no	no	no	no	no	no
Dagbani	not confirmed	no	no	no	no	no

Table 1. Which Ghanaian languages each resource claims. Ga's absence from FLORES-200 was checked against the released file list; Dagbani was not checked against every resource.

Ga is the language of Accra, the country's largest urban area, and none of these resources covers it. Ewe has benchmark files but no social media data, so nothing in this paper's experiments could be repeated for it.

5.2 Documentation audit (RQ1)

Criterion	FLORES-200	AfriSenti	AfriHate	MasakhaNER 2.0
A1 Language referent	Two files, `aka_Latn` and `twi_Latn`; no dialect	"Twi", no dialect	"Twi", no dialect	"Akan/Twi", no dialect
A2 Natively written	No, translated from English	Yes, tweets	Yes, tweets	No, MAFAND-MT translation
A3 Native-speaker annotators	Human-verified translation	Yes	Yes	Yes
A4 Agreement	n/a	κ = 0.51	0.46–0.81 across languages	κ = 0.932, after adjudication
A5 Annotator pay disclosed	Not found	Not found	"more than the minimum wage"	"appropriate remuneration"
A6 Datasheet or data statement	Not found	Not found	Not found	Not found

Table 2. Disclosure matrix. Evidence lines for every cell are in `reports/evidence/`.

Nobody says which Twi. FLORES-200 has two files, `aka_Latn` and `twi_Latn`. They are not copies: only 8 of 2,009 lines are identical, and the Jaccard similarity of their vocabularies is 0.36. MasakhaNER 2.0 merges the two names into "Akan/Twi"; AfriSenti and AfriHate say "Twi". None of the four papers mentions Akuapem, Asante or Fante, the varieties a Ghanaian reader would ask about.

The benchmarks are translations. FLORES-200 is English Wikimedia text translated into Twi. For Twi, MasakhaNER 2.0 uses MAFAND-MT, which its paper describes as "French and English news articles translated into 11 languages". The authors add: "We note that translationese may lead to undesired properties, e.g., unnaturalness. However, we did not observe serious issues during the annotation." That second sentence is about whether annotators could label the text reliably. It says nothing about whether the text resembles Twi as people write it, which is the property at issue here. The only natively written Twi among the four resources is in the two Twitter collections.

The natively written corpus flags its own difficulty. AfriSenti's authors write that "a significant portion of tweets in Twi were ambiguous, making it difficult to categorize sentiment accurately", and report κ = 0.51 for Twi. They also explain that native speakers labelled 2,000 tweets as Twi or not, and a model trained on those labels assigned the rest, which is the setting Caswell et al. (2020) warn about.

Agreement scores cannot be compared. MasakhaNER 2.0's κ = 0.932 for Twi is measured after language coordinators resolved disagreements; the paper says the pre-intervention score "was much lower" and gives one example (Nigerian Pidgin, 0.648 before and 0.966 after). AfriSenti's 0.51 is a free-marginal kappa with no adjudication step described. A reader choosing between the corpora on reported agreement would be comparing two different quantities.

No datasheets. The A6 search found no datasheet, data statement or data card in any of the four papers. AfriHate and MasakhaNER 2.0 disclose that annotators were paid; AfriSenti does not say.

5.3 Distance from social media text (RQ2)

Corpus	Texts	Mean tokens	Tokens with ɛ/ɔ	Texts writing `3` for ɛ	Tokens with c/j/q/v/x/z	Texts with emoji
FLORES-200 `twi_Latn`	2009	25.3	28.5%	0.1%	3.5%	0.0%
FLORES-200 `aka_Latn`	2009	25.8	26.1%	0.1%	3.1%	0.0%
MasakhaNER 2.0 Twi	6056	23.4	34.6%	0.0%	1.7%	0.0%
AfriSenti Twi	4818	9.3	5.6%	42.5%	4.6%	41.2%

Table 3. Surface profile of the three Twi corpora (`reports/metrics/ladder_profile.csv`).

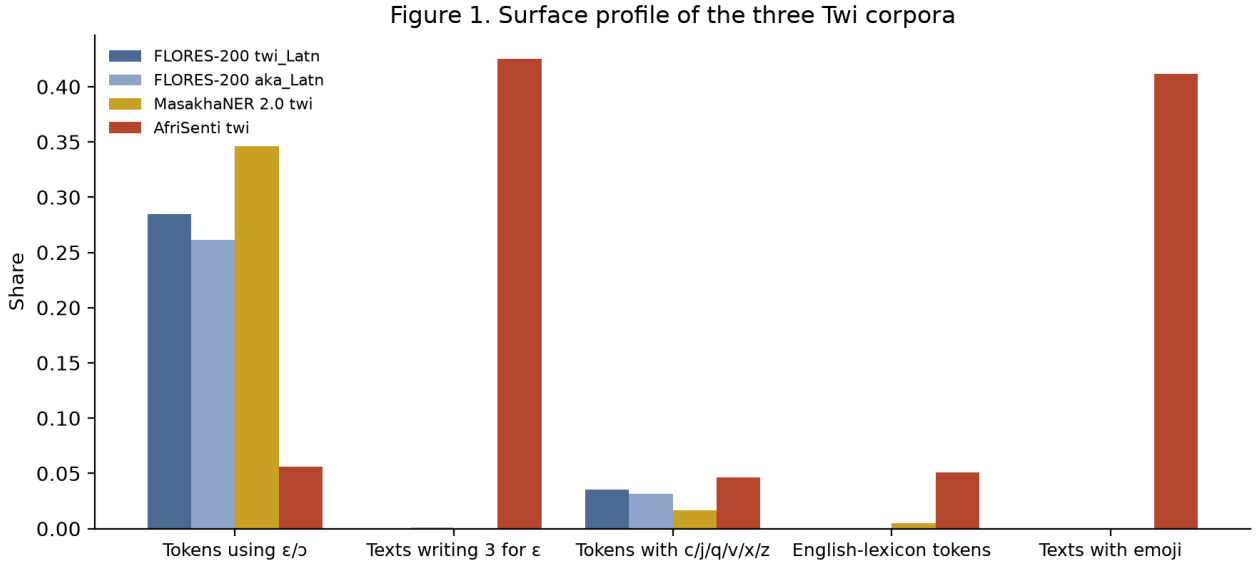


Figure 1. Surface profile of the three Twi corpora. The two benchmarks write ɛ and ɔ; the tweets mostly write 3 or nothing.

The tweets are a different kind of text. They are less than half as long as benchmark sentences. In the benchmarks, between 26.1% and 34.6% of tokens contain ɛ or ɔ; in the tweets, 5.6% do, and 42.5% of tweets write `3` inside a word instead. The English-token estimate is 5.1% for the tweets and 0.5% for MasakhaNER (it is zero for FLORES by construction). Vocabulary overlap is low: FLORES `twi_Latn` covers 15.4% of the word types in the tweets (51.4% of tokens), and MasakhaNER covers 21.9%.

5.4 Sentiment systems on natively written Twi (RQ3)

System	Accuracy	Macro-F1 [95% CI]	F1 neg / neu / pos	Predicted neutral	Silent failure	McNemar p vs baseline
Majority class	0.474	0.214 [0.205, 0.224]	0.00 / 0.00 / 0.64	0.0%	0.0%	< 0.001
TF-IDF + LogReg (baseline)	**0.675**	**0.593** [0.559, 0.625]	0.73 / 0.31 / 0.74	12.4%	9.6%	—
English-only	0.251	0.243 [0.215, 0.271]	0.23 / 0.28 / 0.22	84.2%	83.3%	< 0.001
XLM-T	0.335	0.337 [0.306, 0.367]	0.38 / 0.28 / 0.35	54.8%	53.2%	< 0.001
NLLB `twi_Latn` → English	0.373	0.373 [0.343, 0.403]	0.47 / 0.28 / 0.38	52.1%	50.6%	< 0.001
NLLB `aka_Latn` → English	0.367	0.367 [0.336, 0.396]	0.46 / 0.27 / 0.36	51.5%	50.1%	< 0.001

Table 4. AfriSenti Twi test split, n = 949. Gold neutral rate 15.4%. Silent failure: share of positive or negative tweets predicted neutral. C = 4.0 was chosen on dev.

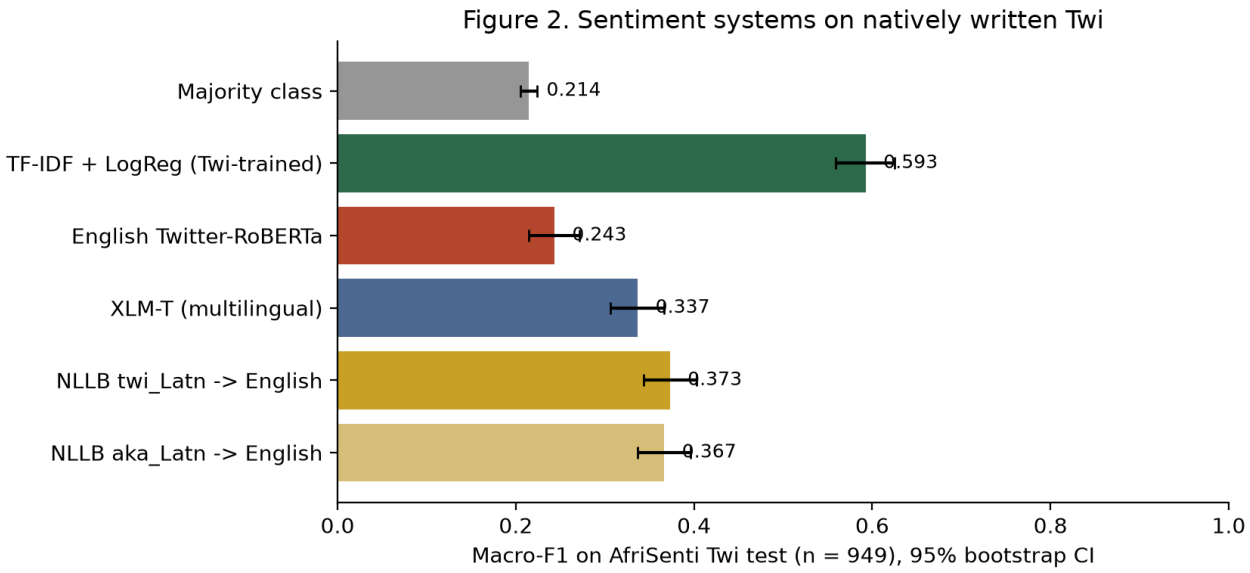


Figure 2. Macro-F1 with 95% bootstrap intervals. Only the model trained on Twi is far from the floor.

The baseline trained on 3,481 Twi tweets reaches macro-F1 0.593. No system that claims wider coverage comes close: translate-then-classify reaches 0.373, XLM-T 0.337, and the English-only model 0.243, just above the majority-class floor of 0.214. Every difference from the baseline is significant (McNemar $p < 0.001$), and none of the confidence intervals overlaps the baseline's.

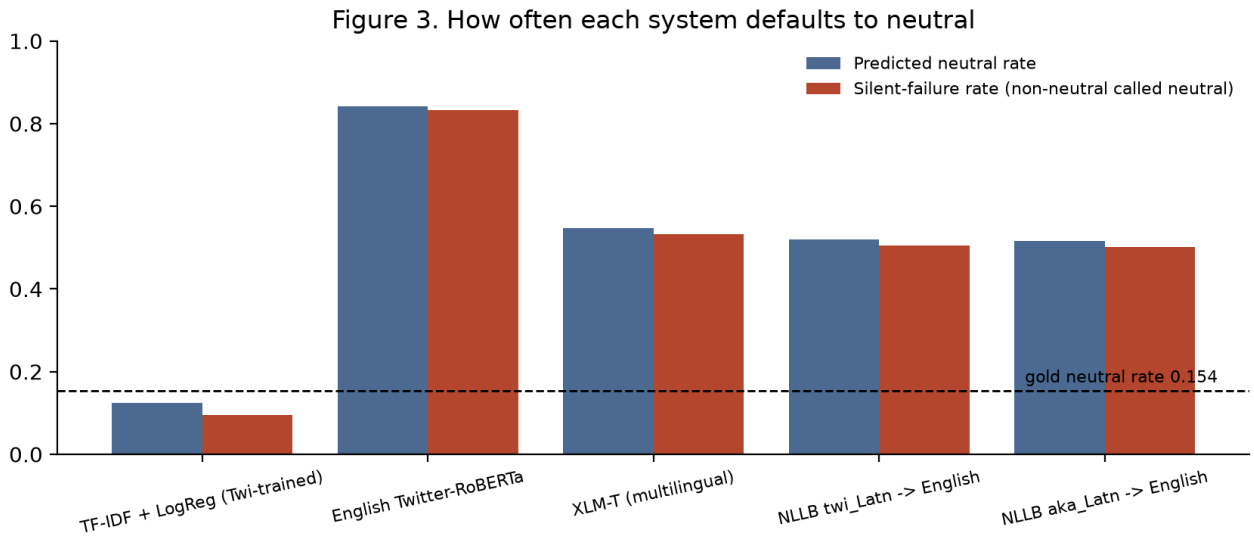


Figure 3. Predicted-neutral and silent-failure rates against the gold neutral rate (dashed).

The failure has a direction. Only 15.4% of test tweets are neutral, yet the English-only model predicts neutral for 84.2% of them and turns 83.3% of the positive and negative tweets into neutral. XLM-T and the NLLB pipeline do the same to about half (53.2% and 50.6%). Averaged over a corpus, this would report Twi speakers as far calmer than they are, and nothing in the output would show it.

6. Error analysis

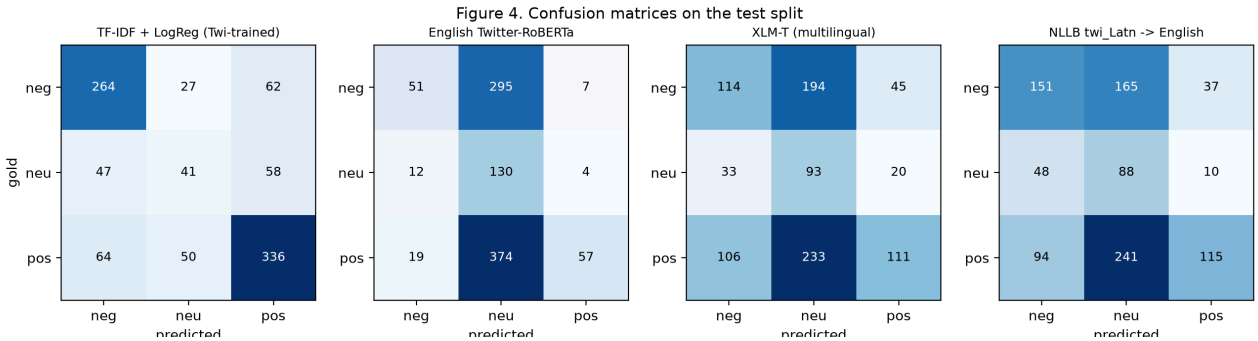


Figure 4. Confusion matrices. The transformer systems pile predictions into the neutral column.

Slice	n	Baseline	English-only	XLM-T	NLLB twi
Uses ɛ/ɔ	215	0.568	0.170	0.264	0.366
Writes `3` for ɛ	380	0.591	0.265	0.326	0.345
Neither	354	0.601	0.263	0.385	0.394
Has emoji	411	0.562	0.362	0.434	0.380
No emoji	538	0.614	0.129	0.239	0.363
English share >25%	26	0.643	0.389	0.492	0.461
No English	651	0.605	0.224	0.324	0.394

Table 5. Macro-F1 by slice (`reports/metrics/slices.csv`; all slices in Figure 5).

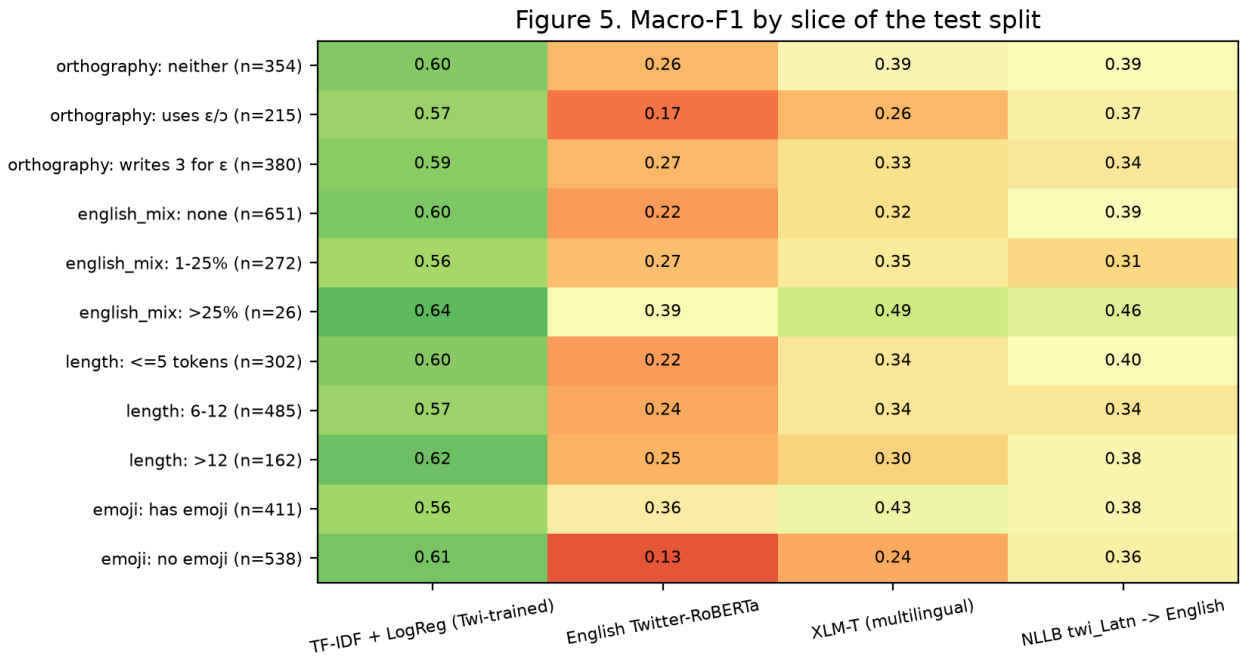


Figure 5. Macro-F1 by slice for the four main systems.

The English model is reading the emoji. On tweets without emoji it predicts neutral for 95.3% of them and scores 0.129; with emoji it scores 0.362. Its errors fall slightly as the English share rises (0.389 above 25%, from 0.224 with none), but that slice holds only 26 tweets.

Translation passes informal spelling through untouched. Of the 387 test tweets that write `3` for ɛ, 188 (48.6%) come back from NLLB with those words still in them: `s3`, `de3`, `y3di` reach the English classifier as unknown strings. Those tweets are then called neutral 60.1% of the time, against 46.7% when the spelling was absorbed. Standard orthography does not rescue translation either: on tweets that use ɛ/ɔ the NLLB pipeline scores 0.366.

Which "Twi" you declare changes the answer. Translating the same tweets as `twi_Latn` and as `aka_Latn` gives identical output for only 120 of 949. The final sentiment label differs for 243 tweets (25.6%), though the two settings score almost the same overall. The naming confusion in Section 5.2 is therefore an input to the result: an analyst who picks the other code gets a different label for one tweet in four, and nothing in the documentation tells them which code is right.

The baseline is steady. Its macro-F1 stays between 0.55 and 0.65 on every slice, including the spelling variants that defeat translation. Its weak point is the neutral class (F1 0.31), the smallest and, given κ = 0.51, probably the noisiest.

Hard cases. 132 tweets are misclassified by the baseline, XLM-T and the NLLB pipeline alike. Samples are saved in `reports/error_analysis/`. I cannot read them well enough to say whether the systems or the gold labels are wrong, and with the creators' own note that many Twi tweets are ambiguous, some of these are likely label errors.

7. Discussion

The three questions have connected answers. The resources that claim Twi measure their claims on translated, standard-orthography prose (RQ1, RQ2), while Twi online is short, mixed with English and emoji, and spelled with digits in place of letters (RQ2). On that text, every system built for broad

coverage loses to a small model trained on 3,481 tweets, and loses by turning sentiment into neutral (RQ3).

For anyone mining Ghanaian social media, the practical conclusion is uncomfortable but cheap to act on. Off-the-shelf multilingual and translation-based pipelines are the wrong default for Twi. Even a modest amount of in-domain labelled data, with a linear model, does better. The pressing work is collecting and labelling more natively written text, for Twi and especially for Ga and Ewe, where none exists.

The documentation findings explain why this is hard to see from the outside. A model card that reports FLORES scores for `twi_Latn` reports performance on translated Wikipedia sentences, in one of two unlabelled Twi files, with no statement of the variety. A practitioner cannot tell from it how the model will behave on tweets, or that switching to `aka_Latn` changes a quarter of the labels.

These teams documented more than most of the field does, and that openness is what made this audit possible. The commercial systems whose claims reach the most users disclose the least, and one of them, Google Translate, could not be tested here at all.

8. Limitations

- **Absent is not the same as not done.** "Not found" in Table 2 means the fact is missing from the public paper. The practice may have happened.
- **Label quality.** Every capability result depends on AfriSenti's labels ($\kappa = 0.51$ for Twi, with ambiguity noted by the authors). Some of what looks like system error is probably label noise.
- **No native-speaker judgement.** I do not read Twi well enough to evaluate translations, so the NLLB results show that sentiment was lost, not that a translation was wrong.
- **One dataset, one task.** Sentiment on one Twitter corpus. Hate speech (AfriHate Twi) and other tasks could behave differently.
- **Models chosen.** XLM-T was not trained on any African language; an Africa-centric model fine-tuned for sentiment might do much better. Google Translate was not tested because I had no API key. Language identifiers (AfroLID, GlotLID) were not run.
- **Surface features are proxies.** The English-token estimate and the orthography slices are heuristics, tested in `tests/`, but not validated against human judgement.
- **Coverage.** Ga's absence was checked against FLORES-200's file list; Dagbani was not checked everywhere. GhanaNLP corpora and the Khaya translator were not audited.

9. Conclusion

Twi is listed as supported by the largest translation systems and by three human-annotated benchmarks. The benchmarks those claims rest on are translations into Twi, written in a spelling that most Twi tweets do not use, under two language codes that nobody distinguishes, and the agreement scores attached to them cannot be compared. On 949 natively written Twi tweets, an English-only model, a multilingual model and a translate-then-classify pipeline all lose to a TF-IDF baseline trained on Twi, mostly by calling emotional tweets neutral.

What is missing is data written the way Ghanaians write, a convention for naming the variety, and agreement reported in comparable form. Until those exist, the safest way to analyse Twi on social media is to label some of it and train on it.

AI-use statement

I used Claude (Anthropic) through Claude Code to brainstorm the topic, write code and draft text. Every quotation was located by script in the source PDFs, every reference was checked against the publisher's record, and every number is written into this manuscript by `scripts/fill_manuscript.py` from files the code produced. Details are in `ai_use_disclosure.md`.

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